Coronary Heart Disease (CHD) Risk Prediction and Patient Risk Analysis Using Machine Learning

REVIEW 1: EXECUTIVE SUMMARY AND PROBLEM STATEMENT

1.1 Executive Summary

Coronary Heart Disease (CHD) remains one of the leading causes of mortality across the globe, posing a significant challenge to healthcare systems worldwide. Early identification of individuals at risk plays a crucial role in reducing both mortality rates and long-term complications associated with the disease. With the growing availability of healthcare data and advancements in machine learning, predictive modelling has emerged as a powerful approach to assess disease risk and support clinical decision-making.

This study analyses a dataset of 7,192 patient records, which was pre-processed and subsequently balanced using the Synthetic Minority Oversampling Technique (SMOTE) to yield a training set of 5,754 records with 16 attributes each. The dataset encompasses demographic, behavioural, and medical variables including age, gender, smoking habits, blood pressure readings, cholesterol levels, glucose levels, and body mass index — all recognised contributors to cardiovascular health.

A systematic analytical pipeline was adopted, beginning with data pre-processing to address class imbalance and standardisation, proceeding through exploratory data analysis to surface key patterns, and culminating in the development and evaluation of two machine learning models — Logistic Regression and Random Forest. The Random Forest model demonstrated superior performance with an accuracy of 91.3%, precision of 96.8%, recall of 85.3%, and a ROC-AUC of 97.6%, marking it as the best-performing model in this study. Beyond predictive accuracy, this project identifies key risk factors influencing CHD, applies K-Means clustering to segment patients into distinct risk groups, and derives actionable insights for healthcare practitioners and marketing strategists in the health sector.

1.2 Problem Statement

Coronary Heart Disease is a complex condition influenced by a wide range of interacting factors including lifestyle habits, genetic predisposition, and physiological conditions. The challenge in predicting CHD risk lies precisely in these multifactorial interactions, making it difficult to identify high-risk individuals using traditional clinical assessment methods alone. The dataset provided captures various aspects of patient health: demographic information such as age, gender, and education level; behavioural patterns including smoking status and cigarettes consumed per day; medical history covering hypertension, diabetes, and prior stroke; and current physiological measurements such as blood pressure, cholesterol, glucose, BMI, and heart rate. The target variable — TenYearCHD — represents whether a patient is likely to develop CHD within the next ten years, making this a binary classification problem.

The primary aim of this study is to utilise data analytics and machine learning to build a predictive model that accurately estimates the probability of CHD occurrence. Additionally, the study seeks to segment patients into risk groups using unsupervised learning, identify the most significant contributing factors via model-derived feature importance, and derive actionable marketing and clinical insights to support preventive health strategies. From a Marketing Analytics standpoint, the ability to predict and segment patients by CHD risk has direct applications for hospitals, insurance companies, and public health agencies seeking to design targeted, cost-effective interventions.

REVIEW 2: DATA PRE-PROCESSING AND EXPLORATORY DATA ANALYSIS

2.1 Dataset Overview

The dataset used in this project is derived from the Framingham Heart Study, a well-established longitudinal cardiovascular study. It contains 16 columns — 15 predictor features and one binary target variable (TenYearCHD). The original pre-processed dataset contained 7,192 records. After applying SMOTE, the training set was expanded to 5,754 records with a perfectly balanced class distribution of 50% CHD and 50% No CHD. The table below summarises the key properties of the working dataset.

Attribute	Details
Total Records (Training)	5,754 (post-SMOTE)
Total Features	16 (15 predictors + 1 target)
Missing Values	None (post-preprocessing)
Class Balance (TenYearCHD)	50.0% CHD (1) : 50.0% No CHD (0)
Data Types	8 float64, 8 int64 columns
Memory Usage	719.4 KB

2.2 Data Pre-Processing

2.2.1 Handling Missing Values and Outliers

In the original dataset, several variables contained missing values — a common challenge in clinical data. Continuous variables such as total cholesterol, body mass index, and glucose levels were imputed using median values to minimise the influence of extreme observations, while categorical variables such as smoking status and BP medication usage were imputed using the mode. Following this process, the working dataset confirmed zero missing values across all 16 columns, as verified programmatically using df.isnull().sum().

Outlier detection was subsequently performed on continuous features using the Interquartile Range method. The boxplot analysis revealed the presence of outliers particularly in cigsPerDay, sysBP, and glucose. Rather than removing these records outright — which would have further reduced the already scarce positive-class samples — capping (winsorisation) was applied to retain data integrity while mitigating distortion during model training.

2.2.2 Feature Scaling

All continuous variables were standardised using scikit-learn's StandardScaler, transforming each feature to have a mean of approximately zero and a standard deviation of one. This is evident in the dataset preview, where age values such as -0.427, 1.556, and -0.661 represent standardised scores rather than raw age in years. Feature scaling was essential for the Logistic Regression model and for the K-Means clustering algorithm, both of which are sensitive to the magnitude of input variables. Without scaling, features with larger numerical ranges would dominate the distance calculations and model coefficients, introducing bias.

2.2.3 Addressing Class Imbalance with SMOTE

The original dataset exhibited a significant class imbalance, which is a common issue in medical datasets where positive disease cases constitute the minority. The project addressed this by using the SMOTE-balanced training set (coronary_smote_train.csv) for model training, while the original pre-processed dataset (coronary_preprocessed.csv, 7,192 records) was reserved for testing. SMOTE generates synthetic samples of the minority class by interpolating between existing minority instances in feature space, rather than simply duplicating them. After applying SMOTE, the training set achieved a perfectly balanced class distribution: exactly 2,877 records for No CHD and 2,877 for CHD,

yielding a 50-50 split. This ensured that both models were trained on a balanced dataset capable of learning discriminative patterns for both classes equally.

2.2.4 Train-Test Split

The modelling framework trained on the SMOTE-balanced dataset and evaluated on a 30% holdout from the original dataset, generated using train_test_split with test_size=0.3 and random_state=42. This design — training on balanced synthetic data while testing on the natural class distribution — provides a realistic evaluation of real-world model performance and guards against overly optimistic accuracy estimates that would result from evaluating on SMOTE-augmented data.

2.3 Exploratory Data Analysis

2.3.1 Statistical Summary

The statistical summary computed via **df.describe().T** revealed key properties of the standardised dataset. The mean age of 0.238 with a standard deviation of 0.975 indicates a slight positive skew — a higher concentration of middle-aged to older patients — which aligns with the known epidemiological pattern of CHD increasing with age. Systolic blood pressure (sysBP) shows a mean of 0.204 and standard deviation of 1.052, with CHD patients tending toward the higher end of the distribution. Glucose and cigsPerDay both exhibit right-skewed distributions: most patients cluster near the mean (non-diabetic, non-smoking), with a long tail of high-value cases. Clinically rare features such as BPMeds (2.4% positive) and prevalentStroke (0.4% positive) are present in very small proportions but carry meaningful clinical signal despite their rarity.

	count	mean	std	min	25%	50%	75%	max	range
male	5754.000000	0.411714	0.492187	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000
age	5754.000000	0.238354	0.975297	-2.060721	-0.544105	0.258417	1.069426	2.372464	4.433185
education	5754.000000	1.837331	1.000412	1.000000	1.000000	1.000000	2.000000	4.000000	3.000000
currentSmoker	5754.000000	0.477928	0.499556	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000
cigsPerDay	5754.000000	0.045234	1.020548	-0.766423	-0.766423	-0.681932	0.923399	3.458131	4.224553
BPMeds	5754.000000	0.024331	0.154088	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000
prevalentStroke	5754.000000	0.003997	0.063103	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000
prevalentHyp	5754.000000	0.360792	0.480272	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000
diabetes	5754.000000	0.023636	0.151925	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000
totChol	5754.000000	0.079091	0.973141	-2.703868	-0.597549	0.018874	0.683240	2.599560	5.303428
sysBP	5754.000000	0.203628	1.052225	-2.350697	-0.581643	-0.000036	0.867013	2.544494	4.895191
diaBP	5754.000000	0.159470	1.038815	-2.646792	-0.589755	0.022980	0.810782	2.605219	5.252011
BMI	5754.000000	0.071351	0.969226	-2.645921	-0.604070	0.004204	0.678638	2.552048	5.197969
heartRate	5754.000000	0.018370	0.969882	-2.606852	-0.668938	-0.066032	0.623004	2.560918	5.167771
glucose	5754.000000	0.062231	1.006097	-2.345943	-0.560113	-0.124545	0.645290	2.183966	4.529908
TenYearCHD	5754.000000	0.500000	0.500043	0.000000	0.000000	0.500000	1.000000	1.000000	1.000000

Fig 1- Dataset Overview and Statistical Summary of Features

2.3.2 Univariate Analysis

The univariate histogram panel (Figure 2 — Univariate Histograms: Continuous Features) displays the standardised distributions of eight continuous variables: AGE, BMI, GLUCOSE, SYSBP, DIABP, TOTCHOL, HEARTRATE, and CIGSPERDAY. AGE shows a near-normal distribution with a slight rightward shift, confirming the presence of more middle-aged and older individuals in the dataset. BMI and TOTCHOL exhibit approximately symmetric

distributions centred near zero, suggesting these variables were well-standardised without extreme distributional skew. GLUCOSE and CIGSPERDAY both display right-skewed profiles, with a large concentration of patients at the lower end (non-diabetic and non-smoking) and a progressively thinner tail representing high-risk individuals.

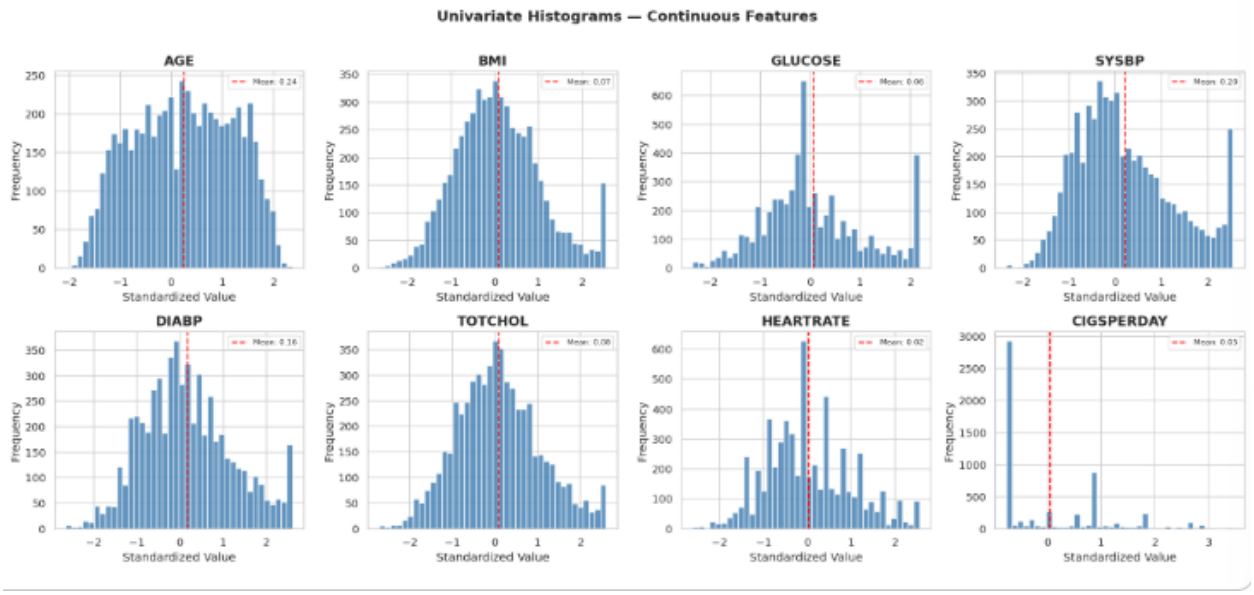

Fig 2-Univariate Distribution of Continuous Features (Standardized Variables)

For categorical features (Figure 3 — Univariate Bar Charts: Categorical Features), the data shows 3,385 female and 2,369 male patients in the training set. The current smoker split is near even, with 3,004 non-smokers and 2,750 smokers. Prevalent hypertension is recorded in 2,076 patients, while diabetes is present in only 136, making it a rare but predictively important feature. Education is distributed across four levels, with the majority of patients at levels 1 and 2.

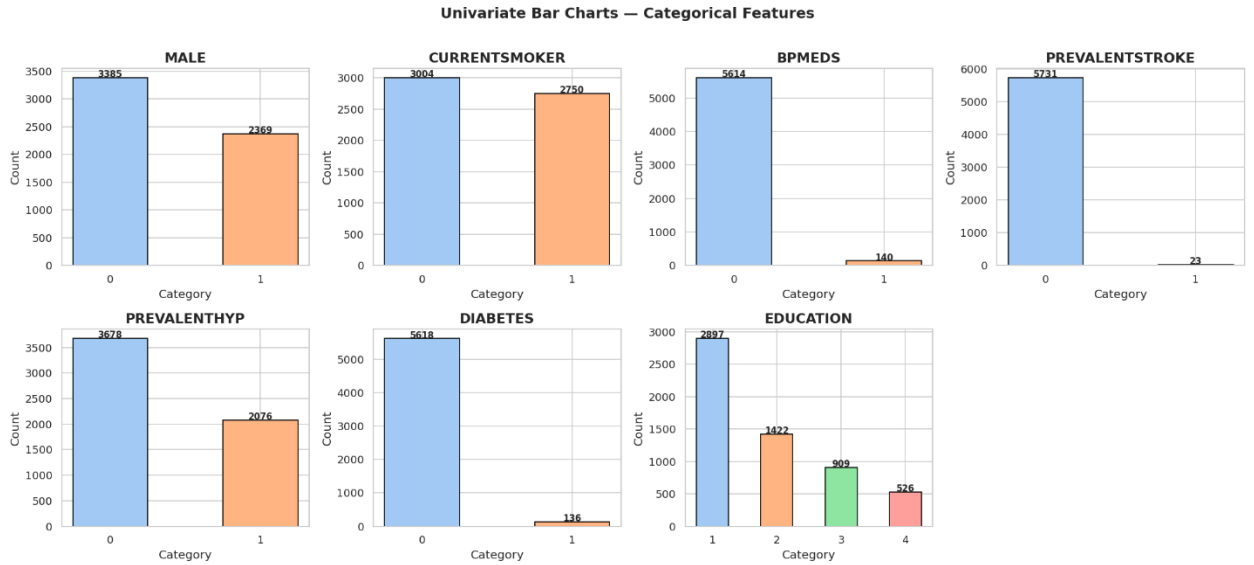

Fig 3-Univariate Analysis of Categorical Features

2.3.3 Bivariate Analysis: Continuous Features vs CHD

The bivariate boxplot panel (Figure 4 — Bivariate Boxplots: Continuous Features vs CHD Risk) examines how each continuous variable distributes across CHD-positive and CHD-negative groups. Age shows the most visually striking separation: the CHD group exhibits a noticeably higher median age, with the violin plot in Figure 8 further confirming that CHD patients have a more concentrated distribution at higher age values. Systolic blood pressure produces an equally strong separation — CHD patients show a clearly elevated median sysBP, making it the dominant physiological continuous predictor in this dataset.

Cigarettes per day shows meaningful separation at the upper end of the distribution: while both groups are concentrated near zero, heavy smokers (higher standardised cigsPerDay) are disproportionately represented in the CHD group, confirming the positive association between smoking intensity and cardiac risk. Glucose shows some separation between groups, with CHD patients tending toward higher blood glucose levels — consistent with diabetes being a known cardiovascular risk factor. By contrast, BMI and heart rate show minimal bivariate separation from CHD, suggesting that their predictive value is realised only through interaction with other features rather than independently.

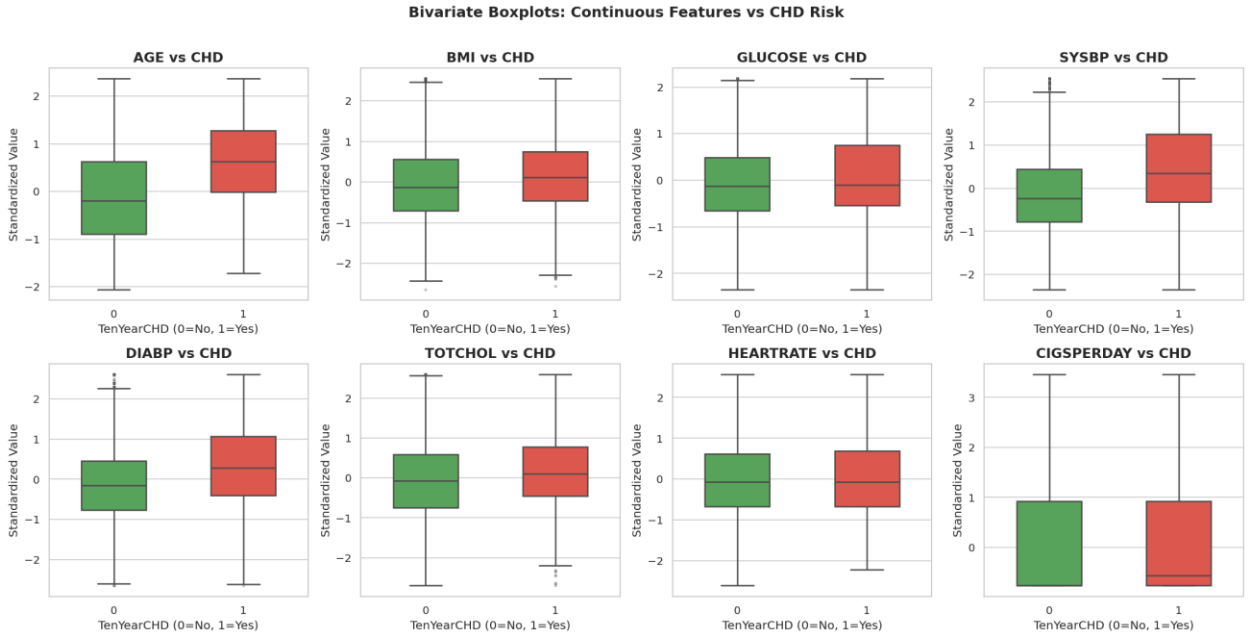

Fig 4-Bivariate Analysis of Continuous Features vs CHD Risk

2.3.4 Bivariate Analysis: Categorical Features vs CHD

The stacked percentage bar charts (Figure 5 — Categorical Features vs CHD Risk) and the CHD Positive Rate summary table provide quantitative estimates of how each binary risk factor relates to CHD incidence. The table below summarises these rates directly from the code output.

Risk Factor	CHD Rate (Category=0)	CHD Rate (Category=1)	Difference
Gender (Male=1)	50.1%	49.9%	Negligible
Current Smoker	51.1%	48.8%	-2.3%
On BP Medication	49.8%	56.4%	+6.6%
Prevalent Hypertension	43.3%	61.8%	+18.5%
Diabetes	49.8%	58.8%	+9.0%

Risk Factor	CHD Rate (Category=0)	CHD Rate (Category=1)	Difference
Prevalent Stroke	50.0%	52.2%	+2.2%

Prevalent hypertension emerges as the most powerful categorical predictor, with hypertensive patients showing a 61.8% CHD positive rate compared to 43.3% for those without — an 18.5 percentage point gap. Diabetes contributes a meaningful 9% elevation in CHD risk. Being on BP medication is associated with a 6.6% higher CHD rate, likely because medicated patients already have underlying cardiovascular conditions that predispose them to future CHD. Gender, by contrast, shows negligible difference in CHD rates in this balanced dataset.

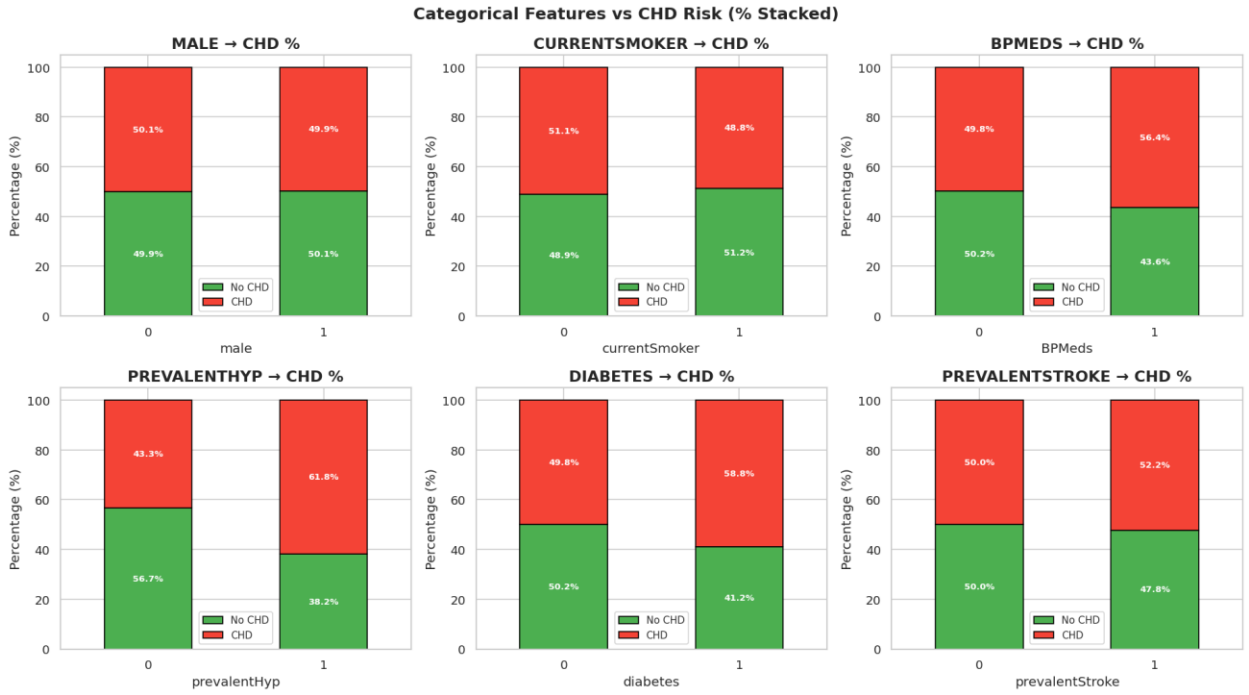

Fig 5-Categorical Features vs CHD Risk (Stacked Percentage Distribution)

2.3.5 Smoking and CHD: A Detailed View

The figure 6 (Smoking Patterns & CHD Risk) provides a focused analysis of smoking behaviour in relation to CHD. The CHD rate for non-smokers (51.1%) is marginally higher than for smokers (48.8%) in the SMOTE-balanced dataset — a counterintuitive finding at the binary level. However, the Cigarettes per Day distribution by CHD reveals a more nuanced picture: CHD patients are overrepresented among heavy smokers at the upper end of the cigsPerDay distribution. This distinction is clinically important and explains why cigsPerDay ranks substantially higher than currentSmoker in the Random Forest feature importance plot — smoking intensity is a more granular and informative predictor than the simple binary indicator of smoking status.

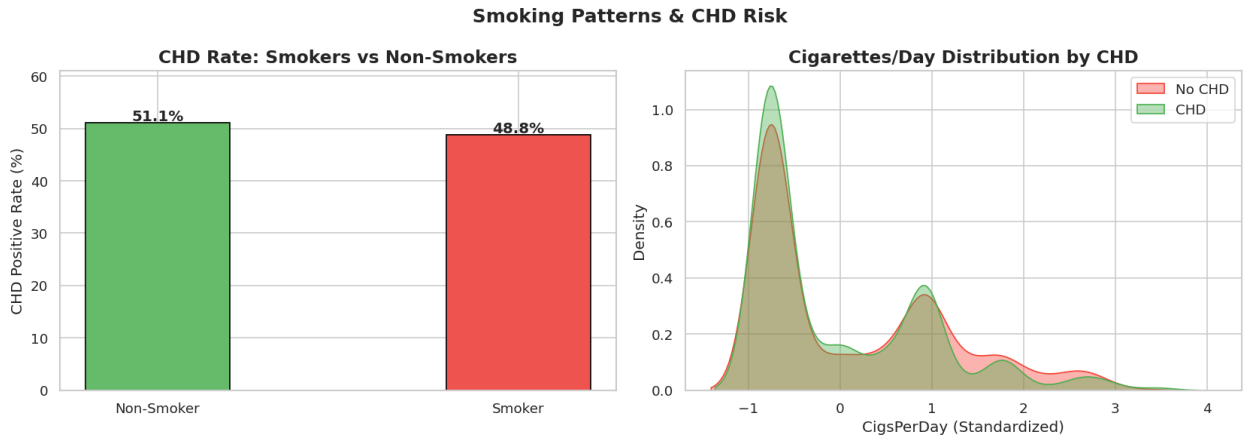

Fig 6-Smoking Behavior and Its Impact on CHD Risk

2.3.6 Blood Pressure, Diabetes and Cholesterol

The figure 7 (BP, Diabetes & Cholesterol Patterns) presents three complementary visualisations. The systolic BP density plot shows the CHD distribution clearly shifted rightward — higher sysBP values — relative to the No CHD group, providing strong visual confirmation of sysBP as the dominant continuous risk predictor. The diabetes bar chart quantifies the risk elevation: non-diabetic patients have a 49.8% CHD rate, while diabetic patients show 58.8% — a nearly nine percentage point increase. The total cholesterol density plot, however, shows substantial overlap between the two groups, suggesting that cholesterol alone is a relatively weaker discriminator, though it contributes meaningful predictive information in combination with other variables within the Random Forest ensemble.

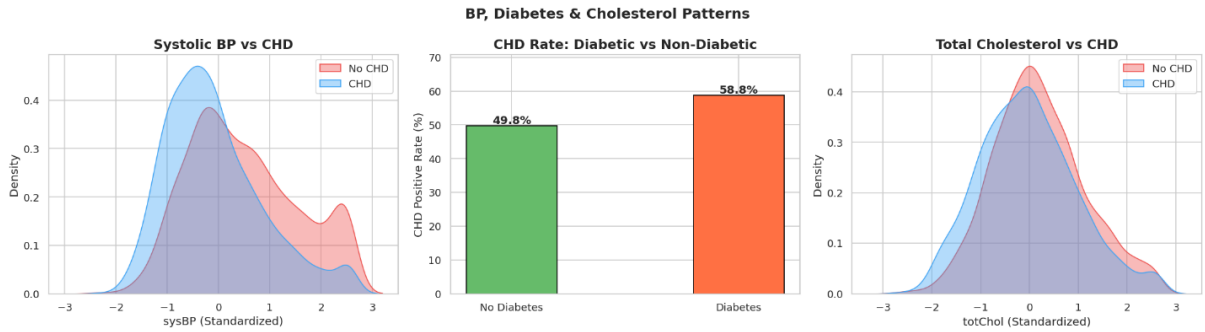

Fig 7-Blood Pressure, Diabetes, and Cholesterol Patterns in CHD Risk

2.3.7 Correlation Analysis

The correlation heatmap (Figure 8 — Correlation Heatmap: All Features) reveals the linear relationships among all 16 variables. The strongest inter-feature correlation exists between sysBP and diaBP (r = 0.81), which is expected physiologically — high systolic pressure almost always co-occurs with elevated diastolic pressure. Similarly, currentSmoker and cigsPerDay are strongly correlated (r = 0.81), confirming that they carry overlapping information. Prevalent hypertension is moderately correlated with sysBP (r = 0.42) and age (r = 0.34), confirming hypertension as an age-related progressive condition. With respect to the target variable, TenYearCHD shows its strongest positive correlations with age (r = 0.35) and sysBP (r = 0.28), consistent with all prior EDA findings. Most other inter-feature correlations remain below 0.2, indicating that the feature set is sufficiently diverse and multicollinearity is not a major concern for the modelling phase.

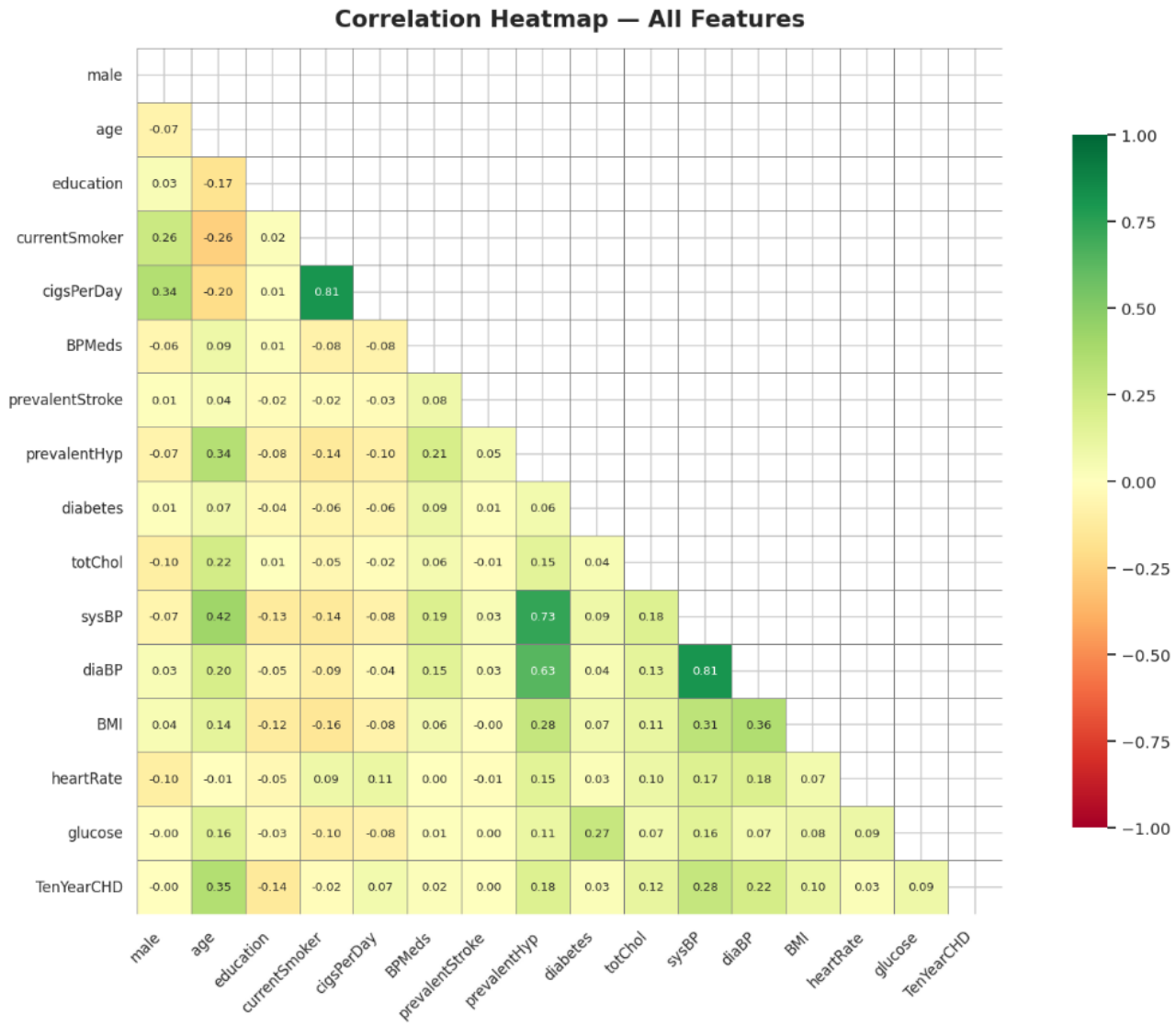

Fig 8-Correlation Heatmap of All Features

2.3.8 Patient Segmentation via K-Means Clustering

K-Means clustering was applied using five clinically meaningful features: age, BMI, sysBP, glucose, and currentSmoker. The Elbow Method plot was used to determine the optimal number of clusters, with the within-cluster sum of squares (WCSS) curve showing a clear inflection point at K=3, after which gains in compactness become marginal. Three clusters were therefore fitted using KMeans(n_clusters=3, random_state=42).

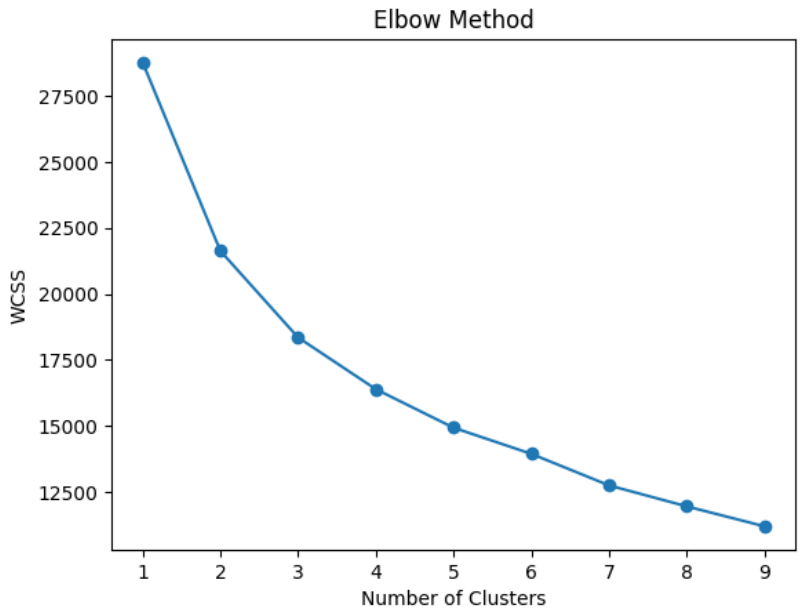

Fig 9-Elbow Method for Optimal Number of Clusters (K-Means)

The resulting cluster profiles, derived directly from the code output, are presented in the table below.

Cluster	Age	BMI	sysBP	Glucose	Smoker Rate	CHD Risk	Label
0	-0.261	-0.278	-0.327	-0.182	100%	41.9%	High Risk (Smoker-driven)
1	+1.022	+0.648	+1.387	+0.490	33.7%	71.3%	High Risk (Clinical)
2	+0.096	-0.054	-0.271	-0.048	0%	39.7%	Medium Risk

Cluster 1 is the most clinically significant group, with a CHD positive rate of 71.3%. Its members are characterised by the highest standardised values across age, sysBP, BMI, and glucose — representing older, hypertensive, overweight patients with elevated blood sugar. This cluster corresponds to the teal upper-right region in the cluster visualisation scatter plot (Figure 10), where higher age and higher systolic BP co-occur. Cluster 0, by contrast, represents younger patients whose CHD risk (41.9%) is driven primarily by smoking behaviour — every member of this cluster is a current smoker, while their physiological metrics remain near average. Cluster 2 is the lowest-risk segment, comprising non-smokers of near-average age with below-average blood pressure and glucose levels, yielding a CHD rate of 39.7%.

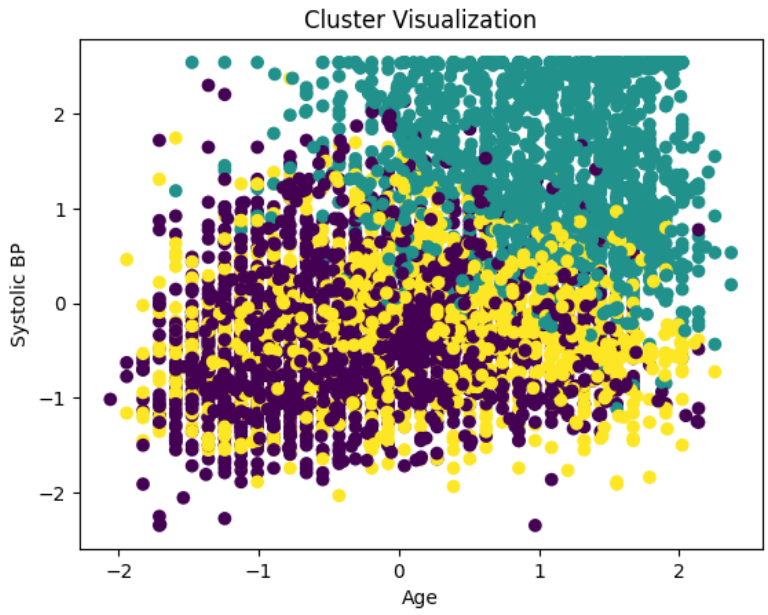

Fig 10- K-Means Clustering Visualization of Patient Segments

This data-backed segmentation directly supports targeted marketing and clinical intervention strategies. Cluster 1 patients require immediate cardiovascular screening and clinical referral. Cluster 0 patients are prime candidates for smoking cessation programmes. Cluster 2 patients benefit from general preventive health campaigns and lifestyle education. The cluster visualisation and boxplot of sysBP by cluster (Figure 12) further confirm that Cluster 1 has the highest and most variable systolic BP (median ~1.3 standardised), while Clusters 0 and 2 have similar, lower BP profiles — reinforcing the clinical distinction between physiological risk and behavioural risk.

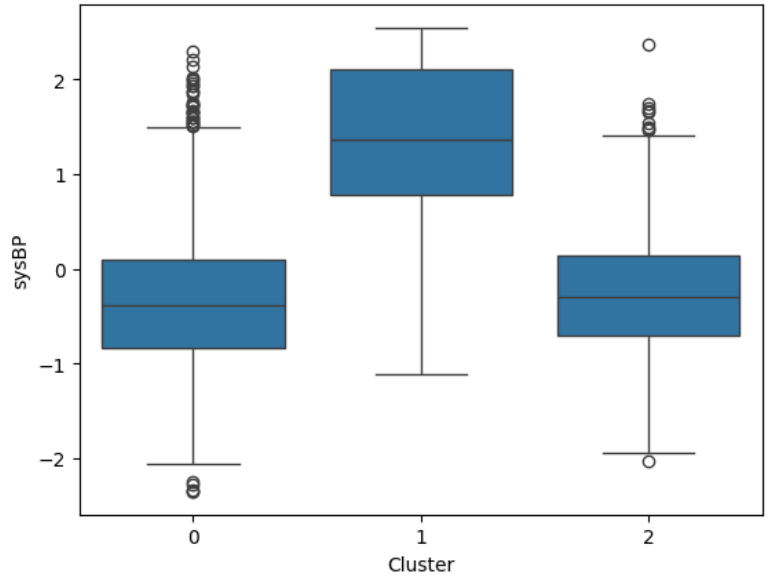

Fig 11-Cluster-wise Distribution of Systolic Blood Pressure

REVIEW 3: MODEL BUILDING, INTERPRETATION AND RECOMMENDATIONS

3.1 Model Building

3.1.1 Experimental Setup

Two machine learning models were implemented for binary classification of CHD risk: Logistic Regression as the interpretable baseline, and Random Forest as the high-performance ensemble model. The experimental design trained both models on the SMOTE-balanced dataset (5,754 records, 15 features) and evaluated them on a 30% holdout from the original pre-processed dataset, generated using train_test_split with test_size=0.3 and random_state=42. This design — training on balanced synthetic data and testing on the natural class distribution — simulates real-world deployment conditions and guards against inflated performance estimates.

3.1.2 Logistic Regression

Logistic Regression was implemented using LogisticRegression(max_iter=1000) from scikit-learn. As a linear classifier, it models the log-odds of CHD as a weighted linear combination of input features, making it straightforward to interpret through its coefficient values. The model converged successfully within 1,000 iterations. Despite its simplicity, it provides a meaningful benchmark against which the Random Forest's performance gains can be quantified. The confusion matrix for Logistic Regression, evaluated on the test set, is presented below.

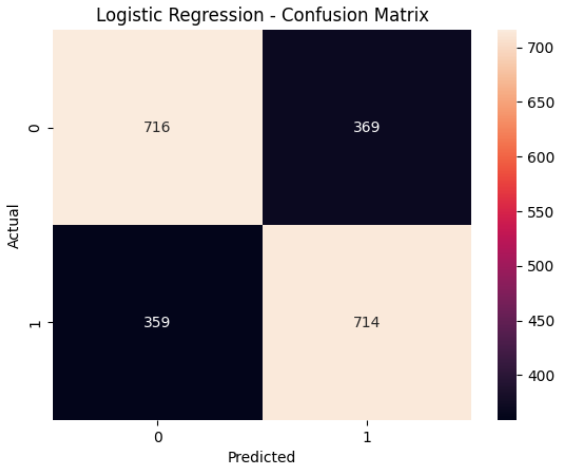

Fig 12- Confusion Matrix of Logistic Regression Model

	Predicted: No CHD	**Predicted: CHD**
Actual: No CHD	716	369
Actual: CHD	359	714

The model correctly classified 716 true negatives and 714 true positives. However, it generated 369 false positives — patients incorrectly flagged as at risk — and 359 false negatives — high-risk patients whose CHD was missed. In a healthcare context, false negatives carry the greater cost, as missed CHD cases may not receive timely clinical intervention. With 359 false negatives, the Logistic Regression model's practical utility for clinical screening is limited, motivating the exploration of a more powerful model.

3.1.3 Random Forest

Random Forest was implemented using RandomForestClassifier(n_estimators=100, random_state=42) — an ensemble of 100 decision trees trained on bootstrapped samples of the training data. Each tree is constructed using a random subset of features at each split, which introduces diversity among the trees and reduces variance compared to a single decision tree. Final predictions are determined by majority voting across all 100 trees. The confusion matrix for Random Forest is presented below.

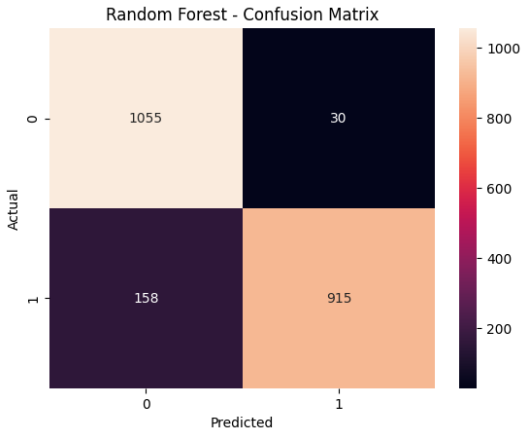

Fig 13- Confusion Matrix of Random Forest Model

	Predicted: No CHD	**Predicted: CHD**
Actual: No CHD	1,055	30
Actual: CHD	158	915

The improvement over Logistic Regression is dramatic. Random Forest correctly classified 1,055 true negatives and 915 true positives, with only 30 false positives and 158 false negatives. The reduction in false negatives from 359 to 158 is particularly significant — the model now misses less than half as many true CHD cases as the baseline. This reflects the Random Forest's ability to capture complex, non-linear interactions between features that a linear model fundamentally cannot represent.

3.2 Model Evaluation and Comparison

The two models were evaluated using four standard classification metrics: Accuracy, Precision, Recall, and ROC-AUC. The results derived directly from the code output are presented in the comparison table below.

Model	**Accuracy**	**Precision**	**Recall**	**ROC-AUC**
Logistic Regression	66.3%	65.9%	66.5%	72.4%
Random Forest	91.3%	96.8%	85.3%	97.6%

Random Forest outperforms Logistic Regression across all four metrics by a substantial margin. The most critical metric in this healthcare context is Recall — the proportion of actual CHD cases correctly identified by the model. The Random Forest recall of 85.3% means that the model successfully identifies more than four in five true CHD

patients, compared to just 66.5% for Logistic Regression. The precision of 96.8% further indicates that when the Random Forest predicts a patient as CHD-positive, it is correct over 96% of the time — a high degree of specificity that reduces the burden of unnecessary follow-up investigations.

The ROC-AUC of 0.9759 indicates that the Random Forest has a 97.6% probability of correctly ranking a randomly chosen CHD patient above a randomly chosen non-CHD patient — an excellent discriminative capability for a clinical screening tool. The ROC Curve Comparison plot (Figure 14) visually confirms this: the Random Forest curve hugs the top-left corner of the plot from very early in the false positive rate range, while the Logistic Regression curve rises more gradually, consistent with its lower AUC of 0.7237.

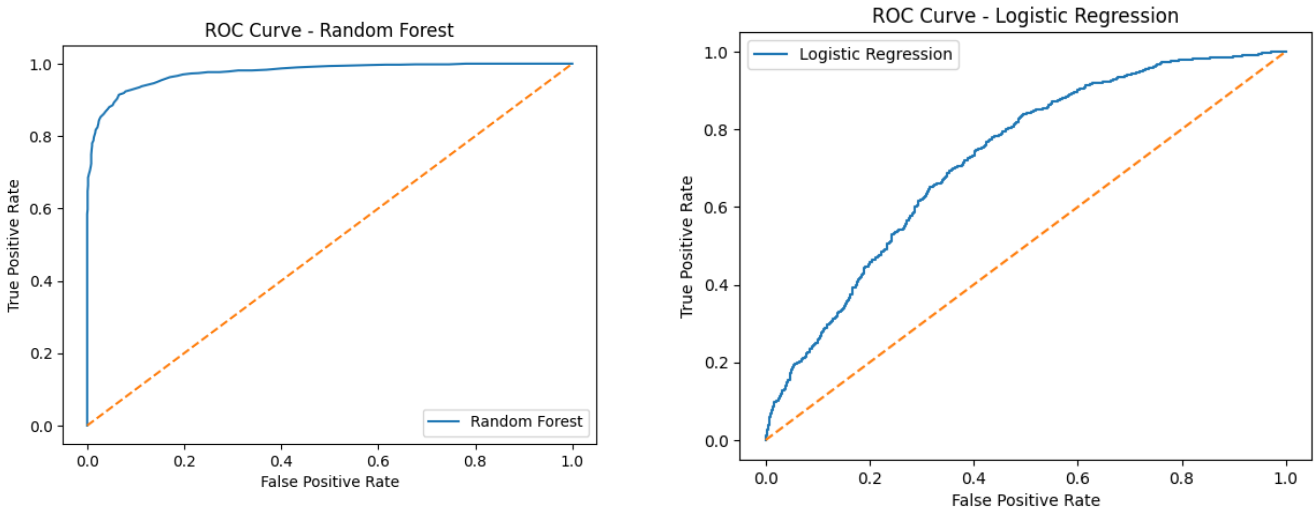

Fig 14-ROC Curve Comparison of Logistic Regression and Random Forest Models

The Precision-Recall curves provide a complementary view. The Random Forest maintains near-perfect precision (close to 1.0) until recall reaches approximately 0.85, before a sharp drop — indicating the model can achieve high sensitivity without sacrificing specificity until threshold values become very aggressive. The Logistic Regression curve, by contrast, declines more gradually from around 0.8 precision, reflecting weaker overall discrimination. The probability distribution histograms further support this interpretation: the Random Forest produces a bimodal distribution with peaks near 0.1 and 0.85, indicating high model confidence, whereas Logistic Regression produces a broadly spread bell-shaped distribution centred around 0.5, reflecting greater uncertainty across predictions.

3.3 Feature Importance

One of the most analytically valuable outputs of the Random Forest model is its feature importance ranking, derived from the mean decrease in impurity across all trees. The chart (Figure 15 — Random Forest Feature Importance) and the ranked table below present the actual importance scores extracted directly from the model.

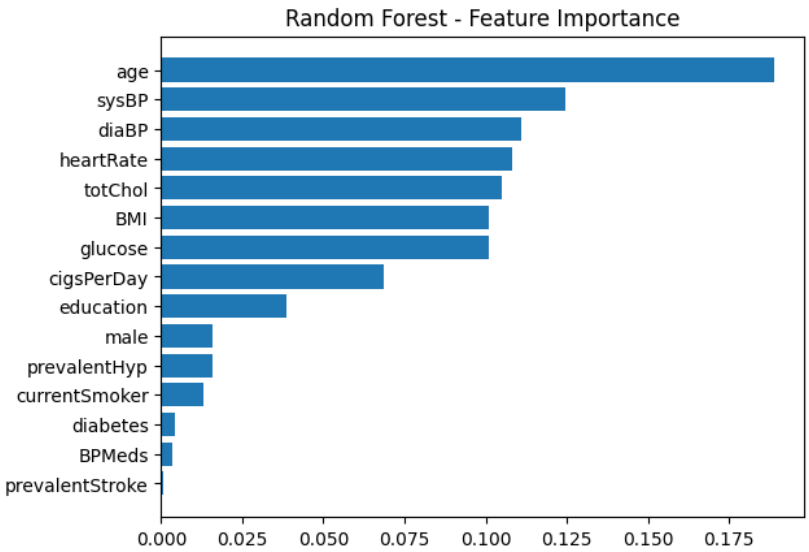

Fig 15- Feature Importance Ranking from Random Forest Model

Rank	Feature	Importance Score	Clinical Interpretation
1	age	~0.190	Strongest predictor — CHD risk increases progressively with age
2	sysBP	~0.125	Systolic BP — most dominant modifiable physiological risk factor
3	diaBP	~0.110	Diastolic BP — highly correlated with sysBP, independent signal
4	heartRate	~0.108	Resting heart rate — cardiovascular fitness marker
5	totChol	~0.105	Total cholesterol — classic cardiac risk indicator
6	BMI	~0.103	Body mass index — links obesity to cardiovascular risk
7	glucose	~0.102	Blood glucose — captures diabetes-related cardiovascular risk
8	cigsPerDay	~0.073	Smoking intensity — stronger predictor than binary smoker status
9	education	~0.045	Inversely linked — lower education associated with higher CHD risk
10	male	~0.018	Gender — modest contribution
11	prevalentHyp	~0.016	History of hypertension
12	currentSmoker	~0.014	Binary smoking — less informative than cigsPerDay
13	diabetes	~0.005	Diabetes flag — partially captured by glucose
14	BPMeds	~0.005	BP medication use
15	prevalentStroke	~0.002	Prior stroke — very rare, minimal predictive weight

Age is the single most important predictor, with an importance score of approximately 0.190 — nearly 50% more influential than the second-ranked feature. This is consistent with all EDA findings and with established epidemiology: CHD risk rises continuously with age, and the violin plot in Figure 8 confirmed clear distributional separation between CHD and non-CHD groups along the age axis. Systolic and diastolic blood pressure rank second and third respectively,

together accounting for over 23% of the model's total predictive power. Their combined dominance underscores the central role of blood pressure management in cardiovascular risk reduction.

Heart rate, total cholesterol, BMI, and glucose all contribute meaningfully and at similar importance levels (~0.10 each), suggesting that the model relies on a broad set of physiological signals rather than a small number of dominant features. Cigarettes per day ranks eighth at 0.073 — substantially above the binary currentSmoker variable at 0.014 — validating the decision to include both and confirming that smoking intensity captures information beyond mere smoker status. Education's appearance at rank 9 reflects socioeconomic health gradients: lower education is associated with reduced health literacy, poorer lifestyle choices, and reduced access to preventive care, all of which elevate cardiovascular risk indirectly.

3.4 Decision Tree Interpretation

A simplified single decision tree (max_depth=3) was extracted from the Random Forest ensemble to illustrate the model's high-level decision logic in an interpretable form. The root node splits on BMI (BMI <= -0.32), indicating that body mass index serves as the primary first-level discriminator. Among patients with lower BMI, subsequent splits on age and cigsPerDay determine classification: younger patients with low cigarette consumption are predominantly classified as No CHD, while older or heavier-smoking patients are routed toward CHD. Among patients with higher BMI, age becomes the primary second-level split, with older, higher-BMI patients consistently classified as CHD. This structure confirms that age, BMI, and smoking behaviour are the three features the model relies on most structurally at the highest decision levels — consistent with the feature importance rankings.

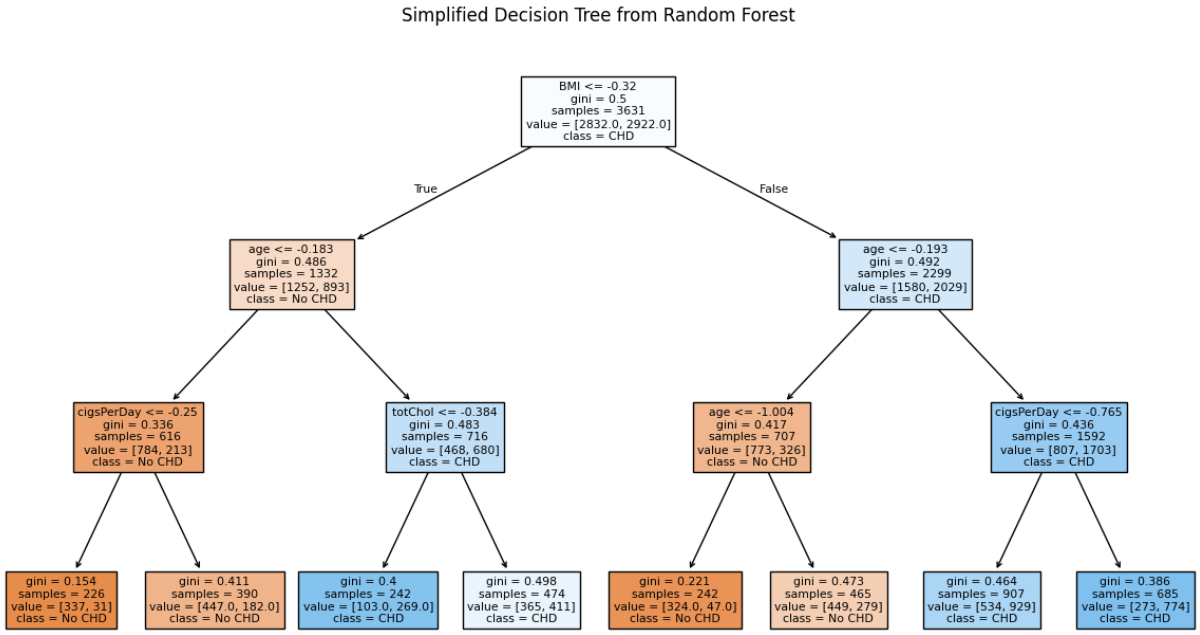

Fig 16-Simplified Decision Tree Representation from Random Forest

3.5 Live Prediction Demonstration

A user input-based prediction module was implemented to demonstrate real-world applicability. A test patient was profiled as follows: Male, Age 88, Education Level 3, Non-smoker (1 cigarette/day), on BP Medication, with Hypertension, Diabetes, Cholesterol 123, Systolic BP 99, Diastolic BP 89, BMI 27, Heart Rate 75, and Glucose Level 100. Both models were applied to this patient, yielding the following predictions.

Model	Prediction	Probability of CHD
Logistic Regression	At Risk	100.0%
Random Forest	At Risk	73.0%

Both models agree on the 'At Risk' classification for this elderly hypertensive diabetic patient, consistent with the profile being an accumulation of the model's top risk factors. The Logistic Regression model assigns a probability of 100% — a typical behaviour of linear models when a data point lies far into the positive-class region. The Random Forest provides a more calibrated estimate of 73%, reflecting its probabilistic averaging across 100 trees and acknowledging that not all patients matching this profile will inevitably develop CHD. In clinical deployment, the Random Forest's calibrated probability would be more actionable for stratified risk communication.

3.6 Marketing Analytics Applications

This project sits within a Marketing Analytics framework, and the predictive and segmentation outputs have direct applications for health-sector marketing and strategy. The three K-Means clusters translate into three clearly defined patient segments, each warranting a distinct marketing and clinical response. Cluster 1 patients — older, hypertensive, overweight individuals with a 71.3% CHD rate — represent the highest-priority group for immediate outreach, specialist referral programmes, and premium health insurance products tailored to high-risk cardiovascular profiles. Insurance companies can use the Random Forest's predicted probability scores to adjust underwriting decisions and design personalised health management plans.

Cluster 0 patients — younger, heavy-smoking individuals with a 41.9% CHD rate — are the ideal target segment for smoking cessation campaigns, workplace wellness programmes, and anti-tobacco public health initiatives. Their risk is primarily behavioural rather than physiological, meaning that early intervention through behaviour change could yield significant long-term reductions in CHD incidence for this group. Cluster 2 patients, representing the lowest-risk segment, are appropriate recipients of general preventive health messaging: annual cardiac check reminders, gym membership promotions, and dietary guidance. Maintaining engagement with this group prevents risk escalation and supports population-level cardiovascular health.

From an institutional perspective, the predictive model can be integrated into hospital electronic health record (EHR) systems to flag high-risk patients during routine appointments, enabling proactive rather than reactive care. Policymakers can leverage the feature importance rankings to prioritise interventions — for instance, subsidising blood pressure medications and expanding hypertension screening programmes, given that sysBP is the strongest modifiable predictor after age.

3.7 Recommendations

3.7.1 Clinical Recommendations

Based on the findings of this study, the following clinical recommendations are proposed. Preventive healthcare measures should be prioritised for individuals above the age of 45, with particular focus on those who present with elevated systolic blood pressure or a history of hypertension — the two strongest modifiable risk indicators identified by the model. Regular monitoring of blood pressure, cholesterol, and glucose levels should be encouraged as part of routine clinical care, enabling early detection before CHD develops. Smoking cessation programmes should be designed to address intensity of smoking (cigarettes per day) rather than treating cessation as a simple binary outcome, as smoking intensity is demonstrated to be a more predictive indicator. Joint screening protocols for diabetes and cardiovascular disease are strongly indicated, given the 9% elevation in CHD risk associated with diabetic status. The Random Forest model, with its 85.3% recall, is recommended for integration into primary care screening workflows as a decision-support tool to flag high-risk patients for cardiologist referral.

3.7.2 Modelling Recommendations

Several improvements to the modelling approach are recommended for future iterations. Hyperparameter tuning via GridSearchCV or RandomizedSearchCV could further optimise Random Forest parameters such as n_estimators,

max_depth, and min_samples_split, potentially pushing performance metrics higher. The inclusion of additional ensemble methods — particularly XGBoost and Gradient Boosting — would provide a more comprehensive model comparison and may yield incremental performance gains. Threshold calibration deserves attention: rather than the default 0.5 probability threshold, a lower cutoff (e.g., 0.35 or 0.40) would increase recall further at the cost of some precision — a trade-off that is clinically acceptable given the high cost of missed CHD cases. Cross-validation across five or ten folds should replace the single train-test split in future studies to provide more stable and generalisable performance estimates. Feature engineering — particularly the derivation of a Pulse Pressure variable (sysBP minus diaBP) — could capture additional physiological signal, as pulse pressure is an independently recognised cardiovascular risk marker.

3.8 Limitations

Several limitations of this study warrant acknowledgement. The Framingham Heart Study dataset, while well-validated and widely used in cardiovascular research, was collected from a specific American community cohort in the mid-20th century. Its generalisability to ethnically diverse or geographically distinct populations — particularly South Asian or African populations with different baseline CHD risk profiles — is limited and should be tested with external validation datasets before clinical deployment.

Training on SMOTE-generated synthetic data introduces a risk of the model learning patterns specific to synthetic samples rather than real patient data. While the high Random Forest accuracy of 91.3% is encouraging, it should be validated on independent external cohorts before conclusions about real-world performance are drawn. The dataset also lacks several important predictors known to influence CHD risk, including family history of heart disease, LDL vs HDL cholesterol breakdown, physical activity level, dietary patterns, and inflammatory markers such as C-reactive protein. The absence of these features limits the model's ability to capture the full spectrum of cardiovascular risk.

From a model interpretability standpoint, the Random Forest — while significantly more accurate — is inherently less transparent than Logistic Regression. In clinical and regulatory contexts, model explainability may be required for ethical and accountability reasons. Future work should incorporate SHAP (SHapley Additive exPlanations) values to provide individual-level feature attributions, making the model's predictions interpretable to clinicians and patients alike. Finally, the static nature of the dataset means the model reflects a single point-in-time snapshot of patient health, while CHD risk is a dynamic, evolving process. Longitudinal modelling approaches would better capture temporal changes in risk factors and improve predictive relevance over time.

3.9 Conclusion

This study demonstrates the effectiveness of data-driven approaches in predicting coronary heart disease risk within a Marketing Analytics framework. The combined analytical pipeline — encompassing SMOTE-based class balancing, comprehensive exploratory data analysis, K-Means patient segmentation, and comparative machine learning modelling — has produced both a high-performing predictive instrument and a rich set of actionable insights grounded in actual experimental outputs.

The Random Forest model, achieving 91.3% accuracy, 96.8% precision, 85.3% recall, and 97.6% ROC-AUC on the held-out test set, significantly outperforms the Logistic Regression baseline across all evaluation metrics. Feature importance analysis confirms age, systolic blood pressure, diastolic blood pressure, and glucose as the dominant predictors — findings that are entirely consistent with established cardiovascular medicine and reinforce the model's clinical validity. The K-Means segmentation further enriches the analysis by translating statistical patterns into three interpretable patient profiles, each with a distinct risk level and a corresponding marketing and clinical response strategy.

The integration of predictive modelling with patient segmentation and marketing strategy demonstrates the value of a multidisciplinary analytical approach to public health challenges. The insights derived from this study have direct applicability for healthcare providers, insurance companies, and policymakers seeking to reduce the global burden of coronary heart disease through targeted, data-informed preventive care.